

Black Hole Simulation: Comparing Hawking Evaporation and Environmental Interaction Models

I developed a Python-based simulation to explore how black holes might evolve under different assumptions. This project compares the standard Hawking radiation model, where black holes gradually lose mass and eventually evaporate, with an alternative model in which black holes interact with surrounding energy.

The purpose of this simulation is not to challenge established physics, but to explore how introducing environmental energy exchange can change the overall behavior of the system. By placing both models side by side, the simulation highlights how different assumptions can lead to very different outcomes.

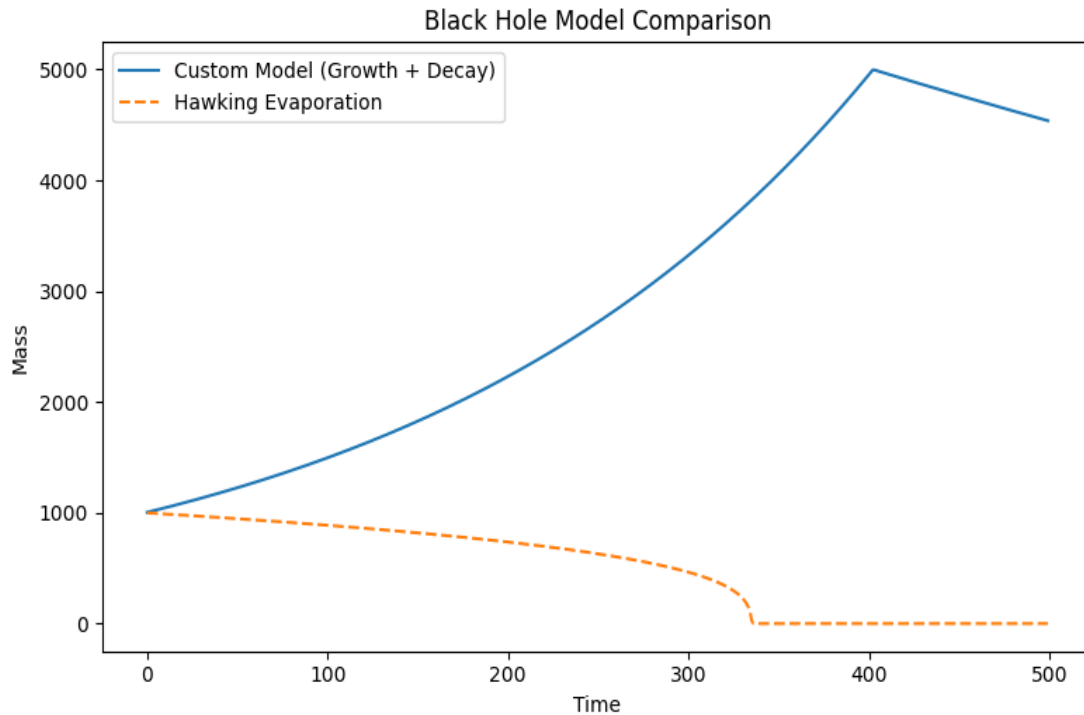
Two models were implemented and compared:

1. Hawking Radiation Model

In the standard model, black holes lose mass over time due to Hawking radiation. As the black hole becomes smaller, the rate of mass loss increases, leading to a rapid final stage of evaporation.

2. Environmental Interaction Model

In the custom model, the black hole is allowed to absorb energy from its surroundings. This creates a dynamic system where the black hole can initially grow in mass, reach a peak, and then begin to decay once the available environmental energy is depleted.



Simulation Results

The graph below shows a comparison between the two models.

Comparison of black hole evolution models. The Hawking model demonstrates continuous mass loss over time, eventually leading to rapid evaporation. In contrast, the environmental interaction model shows three distinct phases: initial growth, a peak phase where energy input slows, and a final decay phase once the surrounding energy is exhausted.

Key Insight

The comparison suggests that black hole behavior may depend strongly on whether interactions with the surrounding environment are considered. While the Hawking model assumes an isolated system, introducing energy exchange produces more complex dynamics, including growth and delayed decay.

This highlights how changes in system assumptions can significantly affect predicted outcomes, even when starting from the same initial conditions.

Limitations

This simulation is a simplified model and does not fully incorporate the complexities of general relativity or quantum field theory. The custom model is not intended to represent a physically validated theory, but rather a conceptual exploration of how alternative assumptions can influence system behavior.

Future improvements could include more physically accurate equations, as well as additional factors such as temperature, spacetime effects, or particle interactions.